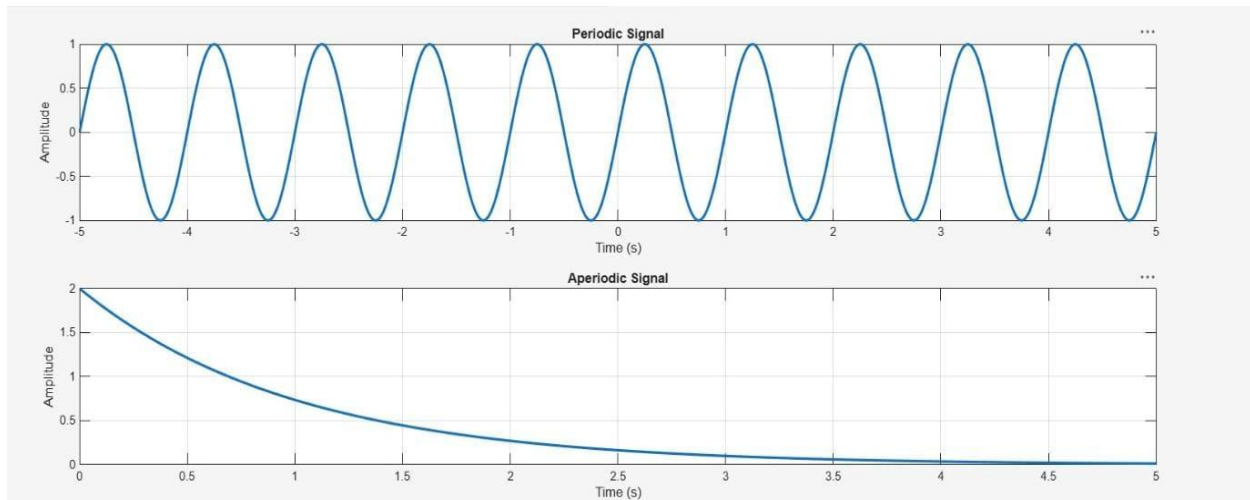


2. SIGNAL PROPERTIES AND TRANSFORMATIONS

Experiment 2.1

Aim : Periodic and Aperiodic Signals



Output

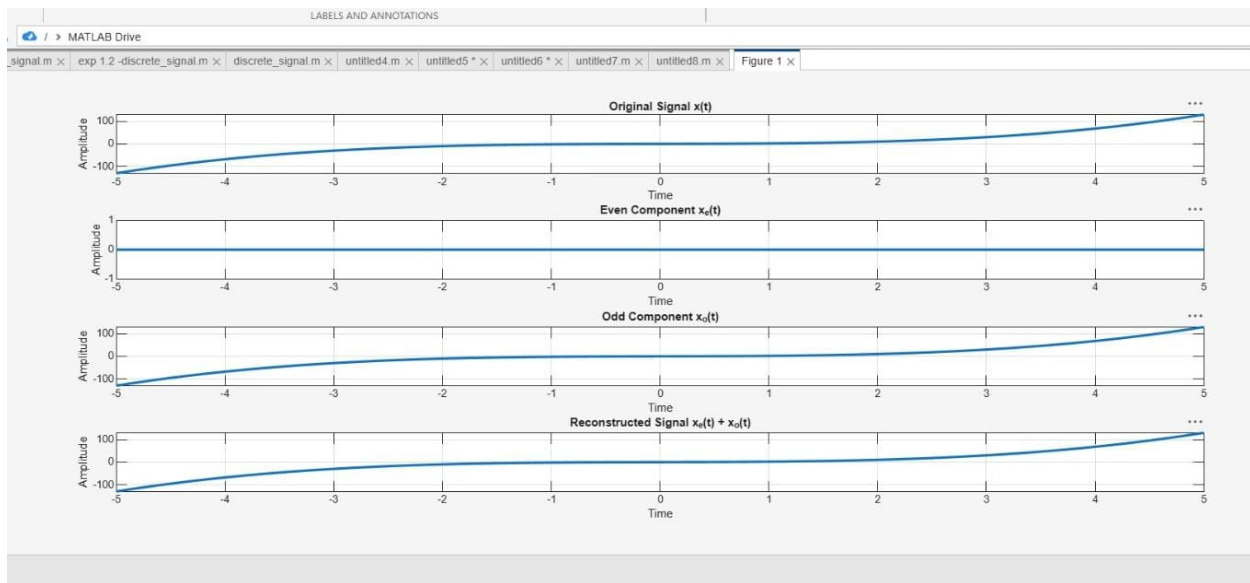
```
1 % Periodic and Aperiodic Signals
2 clc;
3 clear;
4 close all;
5 % Time axis
6 t = -5:0.01:5;
7 % Periodic Signal (Sine Wave)
8 x1 = sin(2*pi*t);
9 % Aperiodic Signal (Exponential Decay)
10 x2 = NaN(size(t));
11 idx = t >= 0;
12 x2(idx) = 2*exp(-t(idx));
13 % Plotting
14 figure;
15 subplot(2,1,1);
16 plot(t,x1,'LineWidth',2);
17 grid on;
18 title('Periodic Signal');
19 xlabel('Time (s)');
20 ylabel('Amplitude');
21 subplot(2,1,2);
22 plot(t,x2,'LineWidth',2);
23 grid on;
24 title('Aperiodic Signal');
25 xlabel('Time (s)');
26 ylabel('Amplitude');
```

Experiment 2.2

Aim : Verification of Even and Odd Signal Decomposition

```
1 % Verification of Even and Odd Signal Decomposition
2 clc;
3 clear;
4 close all;
5 % Time axis
6 t = -5:0.1:5;
7 % Original Signal
8 x = t.^3 + t;
9 % Time-reversed Signal
10 x_rev = (-t).^3 + (-t);
11 % Even Component
12 xe = (x + x_rev)/2;
13 % Odd Component
14 xo = (x - x_rev)/2;
15 % Reconstruction of Original Signal
16 x_reconstructed = xe + xo;
17 % Plotting
18 figure;
19 subplot(4,1,1);
20 plot(t,x,'LineWidth',2);
21 grid on;
22 title('Original Signal x(t)');
23 xlabel('Time');
24 ylabel('Amplitude');
25 subplot(4,1,2);
26 plot(t,xe,'LineWidth',2);
27 grid on;
28 title('Even Component x_e(t)');
29 xlabel('Time');
30 ylabel('Amplitude');
31 subplot(4,1,3);
32 plot(t,xo,'LineWidth',2);
33 grid on;
34 title('Odd component x_o(t)');
35 xlabel('Time');
36 ylabel('Amplitude');
37 subplot(4,1,4);
38 plot(t,x_reconstructed,'LineWidth',2);
39 grid on;
40 title('Reconstructed Signal x_e(t) + x_o(t)');
41 xlabel('Time');
42 ylabel('Amplitude');
```

Output

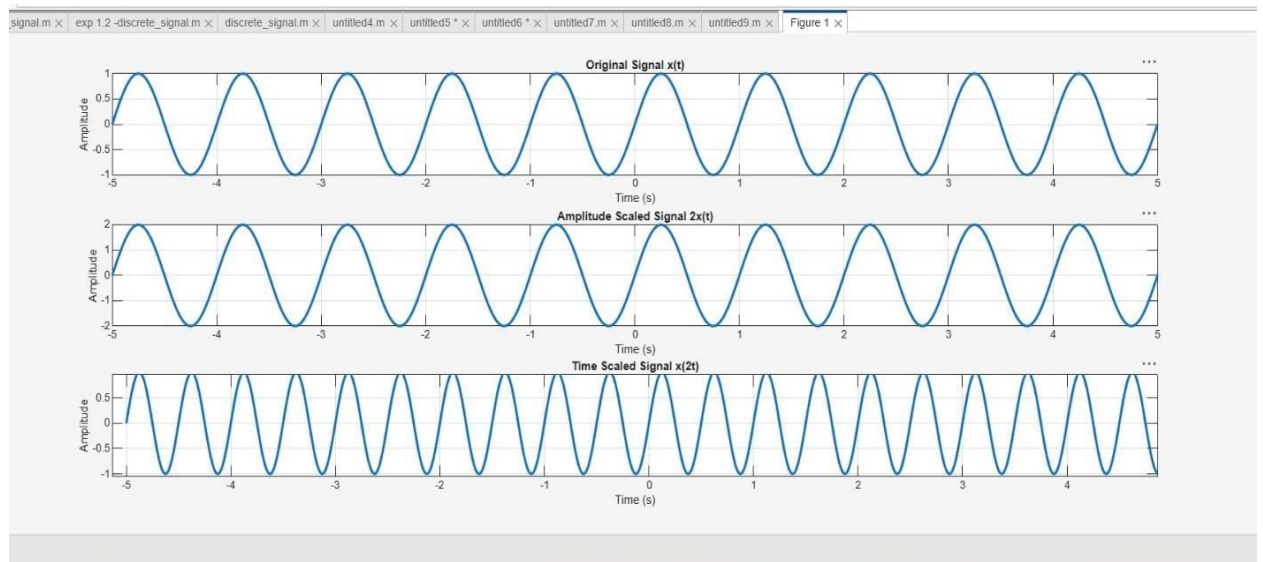


Experiment 2.3

Aim: Amplitude and Time Scaling of Signals

```
1 % Amplitude and Time Scaling of Signals
2 clc;
3 clear;
4 close all;
5 % Time axis
6 t = -5:0.01:5;
7 % Original Signal
8 x = sin(2*pi*t);
9 % Amplitude Scaling
10 x_amp = 2*x;
11 % Time Scaling (Compression)
12 x_time = sin(2*pi*2*t);
13 % Plotting
14 figure;
15 subplot(3,1,1);
16 plot(t,x,'LineWidth',2);
17 grid on;
18 title('Original Signal x(t)');
19 xlabel('Time (s)');
20 ylabel('Amplitude');
21 subplot(3,1,2);
22 plot(t,x_amp,'LineWidth',2);
23 grid on;
24 title('Amplitude Scaled Signal 2x(t)');
25 xlabel('Time (s)');
26 ylabel('Amplitude');
27 subplot(3,1,3);
28 plot(t,x_time,'LineWidth',2);
29 grid on;
30 title('Time Scaled Signal x(2t)');
31 xlabel('Time (s)');
32 ylabel('Amplitude');
```

Output



Experiment 2.4

Aim: Time Shifting and Time Reversal of Signals

```
< exp 1.2 -discrete_signal.m x discrete_signal.m x untitled4.m x untitled5 * x untitled6 * x untitled7.m x untitled8.m x untitled9.m x untitled10.m x Figure 1 x
/MATLAB Drive/untitled10.m
1 % Time Shifting and Time Reversal of Signals
2 clc;
3 clear;
4 close all;
5 % Time axis
6 t = -5:0.01:5;
7 % Original Signal
8 x = sin(2*pi*t);
9 % Time Shift (Delay by 2seconds)
10 x_shift = sin(2*pi*(t-2));
11 % Time Reversal
12 x_reverse = sin(2*pi*(-t));
13 % Plotting
14 figure;
15 subplot(3,1,1);
16 plot(t,x,'LineWidth',2);
17 grid on;
18 title('Original Signal x(t)');
19 xlabel('Time (s)');
20 ylabel('Amplitude');
21 subplot(3,1,2);
22 plot(t,x_shift,'LineWidth',2);
23 grid on;
24 title('Time Shifted Signal x(t-2)');
25 xlabel('Time (s)');
26 ylabel('Amplitude');
27 subplot(3,1,3);
28 plot(t,x_reverse,'LineWidth',2);
29 grid on;
30 title('Time Reversed Signal x(-t)');
31 xlabel('Time (s)');
32 ylabel('Amplitude');
```

Output

